

SOLIDWORKS API (.NET VBA C#) — Reference (193/337): IModelDoc2 (part 16 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 16 of 17) (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchRectangleAtAnyAngle Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchRectangleAtAnyAngle Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::Create3PointCornerRectangle](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchRectangleAtAnyAngle( _  
    ByVal Val1 As System.Double, _  
    ByVal Val2 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal Val3 As System.Double, _  
    ByVal Val4 As System.Double, _  
    ByVal Z2 As System.Double, _  
    ByVal Val3x As System.Double, _  
    ByVal Val3y As System.Double, _  
    ByVal Z3 As System.Double, _  
    ByVal Val5 As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Val1 As System.Double  
Dim Val2 As System.Double  
Dim Z1 As System.Double  
Dim Val3 As System.Double  
Dim Val4 As System.Double  
Dim Z2 As System.Double  
Dim Val3x As System.Double  
Dim Val3y As System.Double  
Dim Z3 As System.Double  
Dim Val5 As System.Boolean
```

```
instance.SketchRectangleAtAnyAngle(Val1, Val2, Z1, Val3, Val4, Z2, Val3x, Val3y, Z3, Val5)
```

C#

```
void SketchRectangleAtAnyAngle(  
    System.double Val1,  
    System.double Val2,  
    System.double Z1,  
    System.double Val3,  
    System.double Val4,  
    System.double Z2,  
    System.double Val3x,  
    System.double Val3y,  
    System.double Z3,  
    System.bool Val5  
)
```

C++/CLI

```
void SketchRectangleAtAnyAngle(  
& System.double Val1,  
& System.double Val2,  
& System.double Z1,  
& System.double Val3,  
& System.double Val4,  
& System.double Z2,  
& System.double Val3x,
```

```

& System.double Val3y,
& System.double Z3,
& System.bool Val5
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val1
X value of corner 1

Val2
Y value of corner 1

Z1
Z value of corner 1

Val3
X value of corner 2 defining the bottom line from corner 1

Val4
Y value of corner 2 defining the bottom line from corner 1

Z2
Z value of corner 2 defining the bottom line from corner 1

Val3x
X value of corner 3; diagonal to corner 1

Val3y
Y value of corner 3; diagonal to corner 1

Z3
Z value of corner 3; diagonal to corner 1

Val5
True to add automatic constraints to the rectangle geometry, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchRectangleAtAnyAngle](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchRectangle Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchSpline Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchSpline Method (IModelDoc2)

Starts a spline, or continues one, using the specified point.

.NET Syntax

Visual Basic (Declaration)

```

Sub SketchSpline( _
    ByVal MorePts As System.Integer, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
)

```

Visual Basic (Usage)

```

Dim instance As IModelDoc2
Dim MorePts As System.Integer
Dim X As System.Double
Dim Y As System.Double

```



```
Dim Z As System.Double
```

```
instance.SketchSpline(MorePts, X, Y, Z)
```

C#

```
void SketchSpline(  
    System.int MorePts,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
void SketchSpline(  
& System.int MorePts,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MorePts
Number of points left to specify the spline after this one (see **Remarks**)
X
x coordinate in meters
Y
y coordinate in meters
Z
z coordinate in meters

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchSpline](#).

Example

[Create 2D Spline \(VBA\)](#)
[Create 2D Spline \(VB.NET\)](#)
[Create 2D Spline \(C#\)](#)

Remarks

MorePts goes from n to 0 in $(n+1)$ calls of this method. When $n = 0$, the last call is made to this method, and the spline is created. For example, if specifying 5 points, then the value of MorePts ranges from 4 to 0.

In SOLIDWORKS 2005 SP1 and earlier, if a failure occurred in a previous call to ModelDoc2::SketchSpline, then subsequent calls to this method failed. In SOLIDWORKS 2005 SP2 and later, if you specify a negative value for MorePts prior to sending the actual data, then the method reinitializes.

In 2D sketches, z is ignored.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISketchSpline Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchSplineByEqnParams Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchSplineByEqnParams Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ISketchSplineByEqnParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchSplineByEqnParams( _  
    ByVal ParamsIn As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ParamsIn As System.Object  
Dim value As System.Boolean  
  
value = instance.SketchSplineByEqnParams(ParamsIn)
```

C#

```
System.bool SketchSplineByEqnParams(  
    System.object ParamsIn  
)
```

C++/CLI

```
System.bool SketchSplineByEqnParams(  
& System.Object^ ParamsIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ParamsIn

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchSplineByEqnParams](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchSplineByEqnParams2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchSplineByEqnParams2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::CreateSplineByEqnParams](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchSplineByEqnParams2( _  
    ByVal ParamsIn As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ParamsIn As System.Object  
Dim value As System.Boolean  
  
value = instance.SketchSplineByEqnParams2(ParamsIn)
```

C#

```
System.bool SketchSplineByEqnParams2(  
    System.object ParamsIn  
)
```

C++/CLI

```
System.bool SketchSplineByEqnParams2(  
& System.Object^ ParamsIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ParamsIn

Array containing an array of doubles (see **Remarks**)

Return Value

True if created successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchSplineByEqnParams2](#).

Example

[Sketch a Spline \(VBA\)](#)

Remarks

The parameters are provided as 3 arrays, which for OLE automation are packed into a single SafeArray packed as follows:

[*Dimension, Order, Number of control Points, Periodicity, knot1, knot2,...,ControlPoint1[Dimension], ControlPoint2[Dimension],...*]

Pass control point coordinates to this method in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your b-curve parameters to sketch space with the help of [ISketch::ModelToSketchTransform](#).

NOTE: If the spline being generated is a closed spline, then it must be flagged as periodic. In addition, splines generated in sketches must be G continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created. If your data is not G1 continuous, then you must split the spline into multiple G1 segments and call method for each segment.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ISketchSplineByEqnParams Method](#)
[IModelDoc2::CreateSplineByEqnParams Method](#)
[IModelDoc2::CreateSplinesByEqnParams Method](#)
[IModelDoc2::ICreateSplineByEqnParams Method](#)
[IModelDoc2::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchTangentArc Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchTangentArc Method (IModelDoc2)

Creates a tangent arc in the current model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchTangentArc()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.SketchTangentArc()
```

C#

```
void SketchTangentArc()
```

C++/CLI

```
void SketchTangentArc();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchTangentArc](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchTrim Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchTrim Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SketchExtend](#) and [ISketchManager::SketchTrim](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchTrim( _  
    ByVal Op As System.Integer, _  
    ByVal SelEnd As System.Integer, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
Dim Op As System.Integer  
Dim SelEnd As System.Integer  
Dim X As System.Double  
Dim Y As System.Double  
  
instance.SketchTrim(Op, SelEnd, X, Y)
```

C#

```
void SketchTrim(  
    System.int Op,  
    System.int SelEnd,  
    System.double X,  
    System.double Y  
)
```

C++/CLI



Copy Code

```
void SketchTrim(  
& System.int Op,  
& System.int SelEnd,  
& System.double X,  
& System.double Y  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Op

SelEnd

X

Y

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchTrim](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchUndo Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchUndo Method (IModelDoc2)

Undoes the last sketch command.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchUndo()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.SketchUndo()
```

C#

```
void SketchUndo()
```

C++/CLI

```
void SketchUndo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchUndo](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ClearUndoList Method](#)
[IModelDoc2::EditUndo2 Method](#)
[IModelDocExtension::FinishRecordingUndoObject Method](#)
[IModelDocExtension::StartRecordingUndoObject Method](#)
[IModelDoc2::EditRedo2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchUseEdge Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchUseEdge Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SketchUseEdge](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchUseEdge()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.SketchUseEdge()
```

C#

```
void SketchUseEdge()
```

C++/CLI

```
void SketchUseEdge();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchUseEdge](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchUseEdge2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchUseEdge2 Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SketchUseEdge](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SketchUseEdge2( _  
    ByVal Chain As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Chain As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SketchUseEdge2(Chain)
```

C#

```
System.bool SketchUseEdge2(  
    System.bool Chain  
)
```

C++/CLI

```
System.bool SketchUseEdge2(  
& System.bool Chain  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Chain

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchUseEdge2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SketchUseEdgeCtrline Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SketchUseEdgeCtrline Method (IModelDoc2)

Uses this centerline in sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub SketchUseEdgeCtrline()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.SketchUseEdgeCtrline()
```

C#

```
void SketchUseEdgeCtrline()
```

C++/CLI

```
void SketchUseEdgeCtrline();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SketchUseEdgeCtrline](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SkToolsAutoConstr Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SkToolsAutoConstr Method (IModelDoc2)

Automatically constrains the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub SkToolsAutoConstr()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.SkToolsAutoConstr()
```

C#

```
void SkToolsAutoConstr()
```

C++/CLI

```
void SkToolsAutoConstr();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SkToolsAutoConstr](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchAddConstraints Method](#)
[IModelDoc2::SketchConstrainCoincident Method](#)
[IModelDoc2::SketchConstrainConcentric Method](#)
[IModelDoc2::SketchConstrainParallel Method](#)
[IModelDoc2::SketchConstrainPerp Method](#)
[IModelDoc2::SketchConstrainTangent Method](#)
[IModelDoc2::SketchConstraintsDel Method](#)
[IModelDoc2::SketchConstraintsDelAll Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SplitClosedSegment Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SplitClosedSegment Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SplitClosedSegment](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SplitClosedSegment( _  
    ByVal X0 As System.Double, _  
    ByVal Y0 As System.Double, _  
    ByVal Z0 As System.Double, _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double _  
)
```

```
Dim instance As IModelDoc2
Dim X0 As System.Double
Dim Y0 As System.Double
Dim Z0 As System.Double
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double

instance.SplitClosedSegment(X0, Y0, Z0, X1, Y1, Z1)
```

C#

```
void SplitClosedSegment(
    System.double X0,
    System.double Y0,
    System.double Z0,
    System.double X1,
    System.double Y1,
    System.double Z1
)
```

C++/CLI

```
void SplitClosedSegment(
    & System.double X0,
    & System.double Y0,
    & System.double Z0,
    & System.double X1,
    & System.double Y1,
    & System.double Z1
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X0
X value of first point
Y0
Y value of first point
Z0
Z value of first point
X1
X value of second point
Y1
Y value of second point
Z1
Z value of second point

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SplitClosedSegment](#).

Remarks

The selected sketch segment must be a closed entity (for example, the start and end points must be the same). To split a closed sketch segment (for example, a complete circle) into two pieces you must specify two points (x1, y1, z1) and (x2, y2, z2).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SplitOpenSegment Method](#)
[ISketchSegment Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SplitOpenSegment Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::SplitOpenSegment](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SplitOpenSegment( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
  
instance.SplitOpenSegment(X, Y, Z)
```

C#

```
void SplitOpenSegment(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
void SplitOpenSegment(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the point that splits the sketch segment in two

Y
Y value of the point that splits the sketch segment in two

Z
Z value of the point that splits the sketch segment in two

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SplitOpenSegment](#).

Remarks

The selected sketch segment must be an open entity; for example, the start and end points cannot be the same.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISketchSegment Interface](#)
[IModelDoc2::SplitClosedSegment Method](#)

Availability

SummaryInfo Property (IModelDoc2)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : SummaryInfo Property (IModelDoc2)

Gets or sets file summary information for the SOLIDWORKS document.

.NET Syntax

Visual Basic (Declaration)

```
Property SummaryInfo( _
    ByVal FieldId As System.Integer _
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim FieldId As System.Integer
Dim value As System.String

instance.SummaryInfo(FieldId) = value

value = instance.SummaryInfo(FieldId)
```

C#

```
System.string SummaryInfo(
    System.int FieldId
) {get; set;}
```

C++/CLI

```
property System.String^ SummaryInfo {
    System.String^ get(System.int FieldId);
    void set (System.int FieldId, System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*FieldId*Identifier for field as defined in [swSummInfoField_e](#)

Property Value

Text in field

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::SummaryInfo](#).

Example

[Get File Summary Information \(VBA\)](#)

Remarks

In line with Microsoft recommendations for OLE support, the file summary information for SOLIDWORKS documents is written as an OLE property set into a stream named "005Summary Information" off the root storage of the SOLIDWORKS document's compound file.

NOTES:

- MFC does not currently provide classes that manage summary information. However, the DRAWCLI application shipped with Visual C++ does include a sample implementation, in the form of the class CSummInfo, which you can use as an example when implementing your own. This class is used by the document class CDrawDoc. DRAWCLI also includes property pages for displaying and modifying Summary Information.
- Some dates are localized.

Option Explicit

```
Sub main()  
    Dim swApp          As SldWorks.SldWorks  
    Dim swModel        As SldWorks.ModelDoc2  
    Dim Text1          As String  
    Dim Text2          As String  
    Dim Text3          As String  
    Dim Text4          As String  
  
    Set swApp = CreateObject("SldWorks.Application")  
    Set swModel = swApp.ActiveDoc  
  
    ' All of these date strings will be localized according to  
    ' the current regional settings  
  
    ' These two dates (day, month, and year)  
    ' will be in numeric format and localized  
    Text1 = swModel.SummaryInfo(swSumInfoCreateDate)  
    Text2 = swModel.SummaryInfo(swSumInfoSaveDate)  
  
    ' These two dates will be the same as displayed in the  
    ' File, Properties menu in text format  
    Text3 = swModel.SummaryInfo(swSumInfoCreateDate2)  
    Text4 = swModel.SummaryInfo(swSumInfoSaveDate2)  
  
    Debug.Print "CreateDate = " + Text1  
    Debug.Print "SaveDate = " + Text2  
    Debug.Print "CreateDate2 = " + Text3  
    Debug.Print "SaveDate2 = " + Text4  
End Sub
```

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::FileSummaryInfo Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Toolbars Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Toolbars Method (IModelDoc2)

Turns the specified SOLIDWORKS toolbars on and off.

Visual Basic (Declaration)

```
Sub Toolbars( _
    ByVal M As System.Boolean, _
    ByVal Vw As System.Boolean, _
    ByVal SkMain As System.Boolean, _
    ByVal Sk As System.Boolean, _
    ByVal Feat As System.Boolean, _
    ByVal Constr As System.Boolean, _
    ByVal Macro As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim M As System.Boolean
Dim Vw As System.Boolean
Dim SkMain As System.Boolean
Dim Sk As System.Boolean
Dim Feat As System.Boolean
Dim Constr As System.Boolean
Dim Macro As System.Boolean
```

```
instance.Toolbars(M, Vw, SkMain, Sk, Feat, Constr, Macro)
```

C#

```
void Toolbars(
    System.bool M,
    System.bool Vw,
    System.bool SkMain,
    System.bool Sk,
    System.bool Feat,
    System.bool Constr,
    System.bool Macro
)
```

C++/CLI

```
void Toolbars(
    & System.bool M,
    & System.bool Vw,
    & System.bool SkMain,
    & System.bool Sk,
    & System.bool Feat,
    & System.bool Constr,
    & System.bool Macro
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

M
True for main toolbar on, false for off
Vw
True for view manipulation toolbar on, false for off
SkMain
RUE for main sketch toolbar on, false for off
Sk
True for sketch entity toolbar on, false for off
Feat
True for feature toolbar on, false for off
Constr
True for relationships toolbar on, false for off
Macro
True for macro toolbar on, false for off

Remarks

See [IModelDoc2::SetToolBarVisibility](#) for control of all SOLIDWORKS toolbars.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetToolBarVisibility Method](#)
[ISldWorks::HideToolBar2 Method](#)
[ISldWorks::GetToolBarState2 Method](#)
[ISldWorks::ShowToolBar2 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ToolsDistance Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ToolsDistance Method (IModelDoc2)

Computes distance.

.NET Syntax

Visual Basic (Declaration)

```
Sub ToolsDistance()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ToolsDistance()
```

C#

```
void ToolsDistance()
```

C++/CLI

```
void ToolsDistance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ToolsDistance](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ToolsGrid Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ToolsGrid Method (IModelDoc2)

Shows and hides the grid in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub ToolsGrid()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ToolsGrid()
```

C#

```
void ToolsGrid()
```

C++/CLI

```
void ToolsGrid();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ToolsGrid](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetGridSettings Method](#)
[IModelDoc2::GridOptions Method](#)
[IDrawingDoc::ToggleGrid Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ToolsMacro Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ToolsMacro Method (IModelDoc2)

Not implemented.

.NET Syntax

Visual Basic (Declaration)

```
Sub ToolsMacro()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ToolsMacro()
```

C#

```
void ToolsMacro()
```

C++/CLI

```
void ToolsMacro();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ToolsMassProps Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ToolsMassProps Method (IModelDoc2)

Calculates the mass properties.

.NET Syntax

Visual Basic (Declaration)

```
Sub ToolsMassProps()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ToolsMassProps()
```

C#

```
void ToolsMassProps()
```

C++/CLI

```
void ToolsMassProps();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ToolsMassProps](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ToolsSketchScale Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ToolsSketchScale Method (IModelDoc2)

Scales a sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub ToolsSketchScale()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ToolsSketchScale()
```

C#

```
void ToolsSketchScale()
```

C++/CLI

```
void ToolsSketchScale();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ToolsSketchScale](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchModifyScale Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ToolsSketchTranslate Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ToolsSketchTranslate Method (IModelDoc2)

Translates a sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub ToolsSketchTranslate()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ToolsSketchTranslate()
```

C#

```
void ToolsSketchTranslate()
```

C++/CLI

```
void ToolsSketchTranslate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ToolsSketchTranslate](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SketchModifyTranslate Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

UnBlankRefGeom Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UnBlankRefGeom Method (IModelDoc2)

Shows the selected, hidden reference geometry in the graphics window.

.NET Syntax

Visual Basic (Declaration)

```
Sub UnBlankRefGeom()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.UnBlankRefGeom()
```

C#

```
void UnBlankRefGeom()
```

C++/CLI

```
void UnBlankRefGeom();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UnBlankRefGeom](#).

Remarks

If you select a reference plane and call this method, then the plane will be visible in the graphics area. This method has the same effect as selecting an item and choosing **Show** on the shortcut menu.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::BlankRefGeom Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

UnblankSketch Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UnblankSketch Method (IModelDoc2)

Shows a hidden sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub UnblankSketch()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.UnblankSketch()
```

C#

```
void UnblankSketch()
```

C++/CLI

```
void UnblankSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UnblankSketch](#).

Example

[Hide and Show Sketches \(C#\)](#)
[Hide and Show Sketches \(VB.NET\)](#)
[Hide and Show Sketches \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::BlankSketch Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UnderiveSketch Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UnderiveSketch Method (IModelDoc2)

Changes a sketch to underived.

.NET Syntax

Visual Basic (Declaration)

```
Sub UnderiveSketch()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.UnderiveSketch()
```

C#

```
void UnderiveSketch()
```

C++/CLI

```
void UnderiveSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UnderiveSketch](#).

Example

[Create Derived or Undersived Sketch \(VB.NET\)](#)
[Create Derived or Undersived Sketch \(VBA\)](#)
[Create Derived or Undersived Sketch \(C#\)](#)

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DeriveSketch Method](#)
[ISketch::IsDerived Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UnLock Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UnLock Method (IModelDoc2)

Reverses [IModelDoc2::Lock](#) and changes the status bar message to **Process Complete**.

 .NET Syntax

Visual Basic (Declaration)

```
Sub UnLock()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.UnLock()
```

C#

```
void UnLock()
```

C++/CLI

```
void UnLock();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UnLock](#).

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UnlockAllExternalReferences Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UnlockAllExternalReferences Method (IModelDoc2)

Unlocks all external references.

 .NET Syntax

Visual Basic (Declaration)

```
Sub UnlockAllExternalReferences()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.UnlockAllExternalReferences()
```

C#

```
void UnlockAllExternalReferences()
```

C++/CLI

```
void UnlockAllExternalReferences();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UnlockAllExternalReferences](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::LockAllExternalReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UnlockFramePoint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UnlockFramePoint Method (IModelDoc2)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Sub UnlockFramePoint()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.UnlockFramePoint()
```

C#

```
void UnlockFramePoint()
```

C++/CLI

```
void UnlockFramePoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UnlockFramePoint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

UserFavors Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UserFavors Method (IModelDoc2)

Specifies whether geometric relations are automatically created as you add sketch entities.

.NET Syntax

Visual Basic (Declaration)

```
Sub UserFavors()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.UserFavors()
```

C#

```
void UserFavors()
```

C++/CLI

```
void UserFavors();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UserFavors](#).

Remarks

If this option is on, then the cursor changes shape as you sketch to show you which relations can be created: horizontal, vertical, parallel, perpendicular, tangent, midpoint, or coincident.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, RevisionNumber 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UserPreferences Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : UserPreferences Method (IModelDoc2)

Obsolete. The **See Also** section of this topic contains links to the methods that supersede this method.

.NET Syntax

Visual Basic (Declaration)

Sub UserPreferences()

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.UserPreferences()
```

C#

```
void UserPreferences()
```

C++/CLI

```
void UserPreferences();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::UserPreferences](#).

Remarks

See [System Options and Document Properties](#) for details.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetUserPreferenceDoubleValue Method](#)
[IModelDoc2::GetUserPreferenceIntegerValue Method](#)
[IModelDoc2::GetUserPreferenceStringValue Method](#)
[IModelDoc2::GetUserPreferenceTextFormat Method](#)
[IModelDoc2::GetUserPreferenceToggle Method](#)
[IModelDoc2::IGetUserPreferenceTextFormat Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

VersionHistory Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : VersionHistory Method (IModelDoc2)

Gets an array of strings indicating the versions in which this document was saved, including the SOLIDWORKS Design version in which the model document is currently opened and which is the last value returned in the array.

.NET Syntax

Visual Basic (Declaration)

```
Function VersionHistory() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Object  
  
value = instance.VersionHistory()
```

C#

```
System.object VersionHistory()
```

C++/CLI

```
System.Object^ VersionHistory();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::VersionHistory](#).

Remarks

There is one array entry for each major release of the SOLIDWORKS Design software in which the document has been saved. The format for each entry is a major release code followed by one or more minor release codes separated by commas:

<major release code>[<minor release code>]

- or -

<major release code>[<minor release code>,<minor release code>...]

where <major release code> equals a number that remains constant through a major release of the SOLIDWORKS Design software. For example, the following values are returned based on the corresponding major SOLIDWORKS Design version:

SOLIDWORKS Release	Version Number
SOLIDWORKS 95	44
SOLIDWORKS 96	243
SOLIDWORKS 97	483
SOLIDWORKS 97Plus	629
SOLIDWORKS 98	822
SOLIDWORKS 98Plus	1008
SOLIDWORKS 99	1137
SOLIDWORKS 2000	1500
SOLIDWORKS 2001	1750
SOLIDWORKS 2001Plus	1950
SOLIDWORKS 2003	2200
SOLIDWORKS 2004	2500
SOLIDWORKS 2005	2800
SOLIDWORKS 2006	3100
SOLIDWORKS 2007	3400
SOLIDWORKS 2008	3800
SOLIDWORKS 2009	4100
SOLIDWORKS 2010	4400
SOLIDWORKS 2011	4700
SOLIDWORKS 2012	5000
SOLIDWORKS 2013	6000
SOLIDWORKS 2014	7000
SOLIDWORKS 2015	8000
SOLIDWORKS 2016	9000
SOLIDWORKS 2017	10000

SOLIDWORKS 2018	11000
SOLIDWORKS 2019	12000
SOLIDWORKS 2020	13000
SOLIDWORKS 2021	14000
SOLIDWORKS 2022	15000
SOLIDWORKS 2023	16000
SOLIDWORKS 2024	17000
SOLIDWORKS 2025	18000
SOLIDWORKS 2026	19000

The <minor release code> equals the year and day of manufacture of a saving version (for example, 1997/320).
To get the version in which a model document was last saved, use [ISldWorks::VersionHistory](#).

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IVersionHistory Method](#)
[ISldWorks::RevisionNumber Method](#)
[IModelDocExtension::IsFutureVersion Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help [Send comments on this topic.](#)

ViewConstraint Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewConstraint Method (IModelDoc2)

Shows the constraints for the current model document.

 .NET Syntax

Visual Basic (Declaration)

```
Sub ViewConstraint()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
instance.ViewConstraint()
```

C#

```
void ViewConstraint()
```

C++/CLI

```
void ViewConstraint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewConstraint](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::DeleteAllRelations Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDispCoordinateSystems Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDispCoordinateSystems Method (IModelDoc2)

Toggles the display of coordinate systems on and off.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDispCoordinateSystems()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewDispCoordinateSystems()
```

C#

```
void ViewDispCoordinateSystems()
```

C++/CLI

```
void ViewDispCoordinateSystems();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDispCoordinateSystems](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDisplayCurvature Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDisplayCurvature Method (IModelDoc2)

Toggles the display of surface curvature on and off.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDisplayCurvature()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewDisplayCurvature()
```

C#

```
void ViewDisplayCurvature()
```

C++/CLI

```
void ViewDisplayCurvature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDisplayCurvature](#).

Example

[Toggle Display of Surface Curvatures \(VBA\)](#)
[Display Hidden Lines \(VBA\)](#)
[Display Hidden Lines \(VB.NET\)](#)
[Display Hidden Lines \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewDisplayFaceted Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDisplayFaceted Method (IModelDoc2)

Sets the current display mode to show the facets that make up a shaded picture of STL output.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDisplayFaceted()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewDisplayFaceted()
```

C#

```
void ViewDisplayFaceted()
```

C++/CLI

```
void ViewDisplayFaceted();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDisplayFaceted](#).

Example

```
Option Explicit

Sub main()

    Dim swApp                As SldWorks.SldWorks
    Dim swModel              As SldWorks.ModelDoc2

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    swModel.ViewDisplayFaceted

End Sub
```

Example

[Display Hidden Lines \(VBA\)](#)
[Display Hidden Lines \(VB.NET\)](#)
[Display Hidden Lines \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewDisplayHiddengreyed Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDisplayHiddengreyed Method (IModelDoc2)

Sets the current display mode to **Hidden Lines Visible**.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDisplayHiddengreyed()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2

instance.ViewDisplayHiddengreyed()
```

C#

```
void ViewDisplayHiddengreyed()
```

C++/CLI

```
void ViewDisplayHiddengreyed();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDisplayHiddengreyed](#).

Example

[Display Hidden Lines \(VBA\)](#)
[Display Hidden Lines \(VB.NET\)](#)

[Display Hidden Lines \(C#\).](#)

Remarks

The hidden lines are shown in grey.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewDisplayHiddenremoved Method](#)
[IModelDoc2::ViewDisplayShaded Method](#)
[IModelDoc2::ViewDisplayWireframe Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDisplayHiddenremoved Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDisplayHiddenremoved Method (IModelDoc2)

Sets the current display mode to **Hidden Lines Removed**.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDisplayHiddenremoved()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewDisplayHiddenremoved()
```

C#

```
void ViewDisplayHiddenremoved()
```

C++/CLI

```
void ViewDisplayHiddenremoved();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDisplayHiddenremoved](#).

Example

[Display Hidden Lines \(VBA\)](#)
[Display Hidden Lines \(VB.NET\)](#)
[Display Hidden Lines \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewDisplayHiddengreyed Method](#)
[IModelDoc2::ViewDisplayShaded Method](#)
[IModelDoc2::ViewDisplayWireframe Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ViewDisplayShaded Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDisplayShaded Method (IModelDoc2)

Sets the current display mode to **Shaded**.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDisplayShaded()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewDisplayShaded()
```

C#

```
void ViewDisplayShaded()
```

C++/CLI

```
void ViewDisplayShaded();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDisplayShaded](#).

Example

[Display Hidden Lines \(VBA\)](#)
[Display Hidden Lines \(VB.NET\)](#)
[Display Hidden Lines \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewDisplayHiddengreyed Method](#)
[IModelDoc2::ViewDisplayHiddenremoved Method](#)
[IModelDoc2::ViewDisplayWireframe Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ViewDisplayWireframe Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDisplayWireframe Method (IModelDoc2)

Sets the current display mode to **Wireframe**.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDisplayWireframe()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewDisplayWireframe()
```

C#

```
void ViewDisplayWireframe()
```

C++/CLI

```
void ViewDisplayWireframe();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDisplayWireframe](#).

Example

[Display Hidden Lines \(VBA\)](#)
[Display Hidden Lines \(VB.NET\)](#)
[Display Hidden Lines \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewDisplayCurvature Method](#)
[IModelDoc2::ViewDisplayFaceted Method](#)
[IModelDoc2::ViewDisplayHiddengreyed Method](#)
[IModelDoc2::ViewDisplayHiddenremoved Method](#)
[IModelDoc2::ViewDisplayShaded Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDispOrigins Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDispOrigins Method (IModelDoc2)

Toggles the display of origins off and on.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDispOrigins()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewDispOrigins()
```

C#

```
void ViewDispOrigins()
```

C++/CLI

```
void ViewDispOrigins();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDispOrigins](#).

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDispRefaxes Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDispRefaxes Method (IModelDoc2)

Toggles the display of reference axis on and off.

 .NET Syntax

Visual Basic (Declaration)

```
Sub ViewDispRefaxes()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewDispRefaxes()
```

C#

```
void ViewDispRefaxes()
```

C++/CLI

```
void ViewDispRefaxes();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDispRefaxes](#).

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IRefAxis Interface](#)
[IModelDoc2::ViewDispRefplanes Method](#)
[IModelDoc2::ViewDispRefPoints Method](#)
[IModelDoc2::ViewDispTempRefaxes Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDispRefplanes Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDispRefplanes Method (IModelDoc2)

Toggles the display of reference planes on and off.

 .NET Syntax

Visual Basic (Declaration)

```
Sub ViewDispRefplanes()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewDispRefplanes()
```

C#

```
void ViewDispRefplanes()
```

C++/CLI

```
void ViewDispRefplanes();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDispRefplanes](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewDispRefaxes Method](#)
[IModelDoc2::ViewDispRefPoints Method](#)
[IModelDoc2::ViewDispTempRefaxes Method](#)
[IRefPlane Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewDispRefPoints Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDispRefPoints Method (IModelDoc2)

Shows and hides the reference points for the current model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDispRefPoints()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewDispRefPoints()
```

C#

```
void ViewDispRefPoints()
```

C++/CLI

```
void ViewDispRefPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDispRefPoints](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IRefPoint Interface](#)
[IModelDoc2::ViewDispRefaxes Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)
[IModelDoc2::ViewDispTempRefaxes Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewDispTempRefaxes Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewDispTempRefaxes Method (IModelDoc2)

Toggles the display of temporary reference axes on and off.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewDispTempRefaxes()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewDispTempRefaxes()
```

C#

```
void ViewDispTempRefaxes()
```

C++/CLI

```
void ViewDispTempRefaxes();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewDispTempRefaxes](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewDispRefaxes Method](#)
[IModelDoc2::ViewDispRefplanes Method](#)
[IModelDoc2::ViewDispRefPoints Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewOglShading Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewOglShading Method (IModelDoc2)

Sets the current display subsystem to use OpenGL.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewOglShading()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewOglShading()
```

C#

```
void ViewOglShading()
```

C++/CLI

```
void ViewOglShading();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewOglShading](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewOrientationUndo Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewOrientationUndo Method (IModelDoc2)

Undoes previous view orientation changes that were made interactively by the user.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewOrientationUndo()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewOrientationUndo()
```

C#

```
void ViewOrientationUndo()
```

C++/CLI

```
void ViewOrientationUndo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewOrientationUndo](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewRotate Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotate Method (IModelDoc2)

Rotates the view of the current model.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotate()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.ViewRotate()
```

C#

```
void ViewRotate()
```

C++/CLI

```
void ViewRotate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotate](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)

[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewRotateminusx Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotateminusx Method (IModelDoc2)

Dynamically rotates the view around x in a negative direction with the current increment.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotateminusx()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewRotateminusx()
```

C#

```
void ViewRotateminusx()
```

C++/CLI

```
void ViewRotateminusx();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotateminusx](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewOrientationUndo Method](#)

[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewRotateminusy Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotateminusy Method (IModelDoc2)

Dynamically rotates the view around y in a negative direction with the current increment.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotateminusy()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewRotateminusy()
```

C#

```
void ViewRotateminusy()
```

C++/CLI

```
void ViewRotateminusy();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotateminusy](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewRotateminusz Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotateminusz Method (IModelDoc2)

Dynamically rotates the view around z in a negative direction with the current increment.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotateminusz()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewRotateminusz()
```

C#

```
void ViewRotateminusz()
```

C++/CLI

```
void ViewRotateminusz();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotateminusz](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ViewRotateplusx Method (IModelDoc2)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotateplusx Method (IModelDoc2)

Rotates the view around x in a positive direction with the current increment.

.NET Syntax

Visual Basic (Declaration)

Sub ViewRotateplusx()

Visual Basic (Usage)

Dim instance As [IModelDoc2](#)

instance.ViewRotateplusx()

C#

void ViewRotateplusx()

C++/CLI

void ViewRotateplusx();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotateplusx](#).

Example

[Rotate Model \(C#\)](#)[Rotate Model \(VB.NET\)](#)[Rotate Model \(VBA\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ViewRotateplusy Method (IModelDoc2)

[See Also](#) [Example](#)



Rotates the view around y in a positive direction with the current increment.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotateplusy()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ViewRotateplusy()
```

C#

```
void ViewRotateplusy()
```

C++/CLI

```
void ViewRotateplusy();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotateplusy](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomToFit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ViewRotateplusz Method (IModelDoc2)

[See Also](#) [Example](#)



Rotates the view around z in a positive direction with the current increment.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotateplusz()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewRotateplusz()
```

C#

```
void ViewRotateplusz()
```

C++/CLI

```
void ViewRotateplusz();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotateplusz](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewRotXMinusNinety Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotXMinusNinety Method (IModelDoc2)

Dynamically rotates the view by negative 90 about x.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotXMinusNinety()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ViewRotXMinusNinety()
```

C#

```
void ViewRotXMinusNinety()
```

C++/CLI

```
void ViewRotXMinusNinety();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotXMinusNinety](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewRotXPlusNinety Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotXPlusNinety Method (IModelDoc2)

Dynamically rotates the view by 90 about x.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotXPlusNinety()
```

```
Dim instance As IModelDoc2  
  
instance.ViewRotXPlusNinety()
```

C#

```
void ViewRotXPlusNinety()
```

C++/CLI

```
void ViewRotXPlusNinety();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotXPlusNinety](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewRotYMinusNinety Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ViewRotYMinusNinety Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ViewRotYMinusNinety Method (IModelDoc2)

Dynamically rotates the view by negative 90 about y.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewRotYMinusNinety()
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2  
  
instance.ViewRotYMinusNinety()
```

C#

```
void ViewRotYMinusNinety()
```

C++/CLI

```
void ViewRotYMinusNinety();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ViewRotYMinusNinety](#).

Example

[Rotate Model \(VBA\)](#)
[Rotate Model \(VB.NET\)](#)
[Rotate Model \(C#\)](#)

Remarks

To increase the speed of dynamic view changes with hidden-edge display or hidden-grey display, use [IModelView::StartDynamics](#) and [IModelView::StopDynamics](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ViewOrientationUndo Method](#)
[IModelDoc2::ViewRotate Method](#)
[IModelDoc2::ViewRotateminusx Method](#)
[IModelDoc2::ViewRotateminusy Method](#)
[IModelDoc2::ViewRotateminusz Method](#)
[IModelDoc2::ViewRotateplusx Method](#)
[IModelDoc2::ViewRotateplusy Method](#)
[IModelDoc2::ViewRotateplusz Method](#)
[IModelDoc2::ViewRotXMinusNinety Method](#)
[IModelDoc2::ViewRotXPlusNinety Method](#)
[IModelDoc2::ViewRotYPlusNinety Method](#)
[IModelDoc2::ViewZoomin Method](#)
[IModelDoc2::ViewZoomout Method](#)
[IModelDoc2::ViewZoomTo2 Method](#)
[IModelDoc2::ViewZoomtofit2 Method](#)
[IModelDoc2::ViewZoomToSelection Method](#)
[IModelDoc2::ViewZoomto Method](#)
[IModelDocExtension::ViewZoomToSheet Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0